DU Mathsoc Problem Solving: Set 7

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∼Integration Beginners Paper∼

For any assistance or inquiries about the problem sets, please email me at: quizmaster.maths@csc.tcd.ie.

Problem 1. Compute $\displaystyle\int_0^1 \frac{\ln x}{(1+x)^2}dx$.

Problem 2. Compute $\displaystyle\int_0^1 \frac{x \arctan x}{(1+x^2)^2}dx$.

Problem 3. Compute $\displaystyle\int_0^{\frac{\pi}{2}} \frac{x}{1+\sin 2x}dx$.

Problem 4. Compute $\displaystyle\int_0^1 \frac{x^2+1}{x^4+1}dx$.

Problem 5. Compute $\displaystyle\int_0^{\frac{\pi}{2}} \ln(\sin x)dx$.